SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image
A grayscale image appears dull with very little difference between object and background intensities.
(a) Clearly interpret the scenario and explain the cause of low contrast.
(b) Identify all key issues affecting image visibility.
(c) Apply an appropriate enhancement technique to address the problem.
(d) Justify your choice with logical reasoning.
(e) Present your explanation in a clear and structured manner.

2. Uneven Illumination
An image captured under non-uniform lighting shows bright and dark patches.
(a) Interpret the scenario and explain the impact of uneven illumination.
(b) Identify key factors causing intensity variation.
(c) Apply a suitable enhancement method to normalize illumination.
(d) Justify your selected approach with reasoning.
(e) Provide a clear and well-structured explanation.

3. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.
(b) Identify key issues impacting image quality.
(c) Apply an appropriate filtering technique.
(d) Justify why this method is more suitable than alternatives.
(e) Present your response clearly and logically.

4. Blurred Edges
An image processed for noise removal appears overly smooth, causing edges to blur.
(a) Interpret the scenario and explain why edges are blurred.
(b) Identify key issues related to smoothing and edge loss.
(c) Apply an appropriate technique to restore edges.
(d) Justify your decision with clear reasoning.
(e) Structure your explanation clearly.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) Interpret the scenario and describe high-frequency noise.
(b) Identify the key issues affecting image clarity.
(c) Apply an appropriate frequency domain filtering method.
(d) Justify the choice of filter with reasoning.
(e) Present your answer in a clear and organized manner.

6. Object Boundary Detection
In a medical image, boundaries between tissues need to be clearly identified.
(a) Interpret the scenario and explain the importance of boundary detection.
(b) Identify key challenges in detecting boundaries.
(c) Apply a suitable edge detection method.
(d) Justify your selection with logical reasoning.
(e) Present your explanation clearly and systematically.

7. Simple Segmentation
An image contains objects clearly distinguishable from the background based on intensity.
(a) Interpret the scenario and explain segmentation requirements.
(b) Identify key features enabling separation.
(c) Apply a suitable segmentation technique.
(d) Justify your decision logically.
(e) Present your response clearly.

8. Complex Segmentation
An image contains multiple regions with overlapping intensity values.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues such as intensity overlap.
(c) Apply an appropriate advanced segmentation method.
(d) Justify how the method handles complexity.
(e) Provide a structured and clear explanation.

9. Broken Edges
After applying edge detection, object boundaries appear discontinuous.
(a) Interpret the scenario and explain why edges are broken.
(b) Identify key issues affecting boundary continuity.
(c) Apply a suitable method to improve edge connectivity.
(d) Justify your approach logically.
(e) Present your explanation clearly.

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.
(a) Interpret the scenario and define the combined requirement.
(b) Identify key issues involving noise and edge preservation.
(c) Apply an appropriate sequence of techniques.
(d) Justify the order and selection with strong reasoning.
(e) Present your response in a clear and structured format.

11. Low Brightness Image

An image captured in low light appears very dark with poor visibility.
(a) Interpret the scenario and explain the cause of low brightness.
(b) Identify key issues affecting image visibility.
(c) Apply an appropriate enhancement technique.
(d) Justify your choice with reasoning.
(e) Present your explanation clearly.

12. Overexposed Image

An image appears excessively bright, losing important details.
(a) Interpret the scenario and explain overexposure.
(b) Identify key issues affecting intensity representation.
(c) Apply a suitable correction technique.
(d) Justify your choice logically.
(e) Present your explanation in a structured manner.

13. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.
(a) Interpret the scenario and describe Gaussian noise.
(b) Identify key issues affecting quality.
(c) Apply an appropriate smoothing filter.
(d) Justify your method over alternatives.
(e) Present your explanation clearly.

14. Speckle Noise

An image obtained from ultrasound imaging shows speckle noise.
(a) Interpret the scenario and explain speckle noise characteristics.
(b) Identify key issues affecting clarity.
(c) Apply a suitable filtering technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

15. Loss of Fine Details

An image loses fine details after processing.
(a) Interpret the scenario and explain detail loss.
(b) Identify key issues related to filtering.

(c) Apply a suitable enhancement technique.
(d) Justify your method with reasoning.
(e) Present your explanation clearly.

16. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.
(a) Interpret the scenario and explain the need for edge enhancement.
(b) Identify key issues in detecting edges.
(c) Apply an appropriate method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

17. Background Noise Removal

An image contains background noise affecting segmentation accuracy.
(a) Interpret the scenario and explain noise impact.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable preprocessing method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

18. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.
(a) Interpret the scenario and explain the challenge.
(b) Identify key issues in threshold selection.
(c) Apply a suitable thresholding method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

19. Texture-Based Segmentation

An image contains regions distinguishable by texture rather than intensity.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues with intensity-based methods.
(c) Apply an appropriate segmentation approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

20. Multiple Object Detection

An image contains multiple objects with similar intensity values.
(a) Interpret the scenario and explain segmentation difficulty.
(b) Identify key issues in distinguishing objects.
(c) Apply a suitable segmentation technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

21. Edge Noise Problem

Edges detected are affected by noise.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in edge detection.
(c) Apply a suitable solution.
(d) Justify your method logically.
(e) Present your explanation clearly.

22. Region Merging Requirement

An image contains fragmented regions that need merging.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region segmentation.
(c) Apply an appropriate technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

23. Over-Segmentation

An image is divided into too many regions after segmentation.
(a) Interpret the scenario and explain over-segmentation.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable correction method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

24. Under-Segmentation

Important regions are merged incorrectly during segmentation.
(a) Interpret the scenario and explain under-segmentation.
(b) Identify key issues affecting separation.
(c) Apply a suitable solution.
(d) Justify your approach logically.
(e) Present your explanation clearly.

25. Boundary Refinement

Detected boundaries are rough and inaccurate.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in boundary detection.
(c) Apply a suitable refinement method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

26. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.
(a) Interpret the scenario and explain frequency-based processing.

(b) Identify key issues in spatial domain methods.
(c) Apply an appropriate frequency domain technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

27. Mixed Noise Scenario

An image contains both Gaussian and impulse noise.
(a) Interpret the scenario and explain challenges.
(b) Identify key issues affecting filtering.
(c) Apply a suitable combined approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

28. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.
(a) Interpret the scenario and explain the requirement.
(b) Identify key issues in filtering.
(c) Apply a suitable technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

29. Region Growing Failure

Region growing fails due to incorrect seed selection.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region growing.
(c) Apply a suitable correction.
(d) Justify your method logically.
(e) Present your explanation clearly.

30. Industrial Inspection Enhancement

An industrial system requires detecting defects in noisy images.
(a) Interpret the scenario and define requirements.
(b) Identify key issues in detection.
(c) Apply an appropriate enhancement and segmentation pipeline.
(d) Justify your sequence logically.
(e) Present your explanation clearly.

1. Low Contrast Image

(a) Interpretation

A low-contrast grayscale image has very little difference between the object and the background. The image looks dull because most pixel values are concentrated within a narrow intensity range.

(b) Key Issues

- Poor visibility of objects
- Loss of image details
- Difficult to distinguish foreground from background
- Reduced image quality

(c) Enhancement Technique

Histogram Equalization

(d) Justification

Histogram Equalization spreads the intensity values across the entire range (0–255), increasing contrast and making hidden details more visible.

(e) Structured Explanation

A low-contrast image appears faded because of a limited intensity range. Applying Histogram Equalization improves the contrast by redistributing pixel intensities, making objects clearer and easier to analyze.

2. Uneven Illumination

(a) Interpretation

The image contains bright and dark areas due to non-uniform lighting during image capture.

(b) Key Issues

- Uneven brightness
- Poor visibility in dark regions
- Difficulty in segmentation
- Inconsistent intensity levels
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(c) Enhancement Technique

Adaptive Histogram Equalization (CLAHE)

(d) Justification

CLAHE enhances local contrast instead of the whole image, making both bright and dark regions clearly visible without over-enhancing noise.

(e) Structured Explanation
Uneven illumination causes some image regions to appear brighter than others. CLAHE improves local contrast, producing a uniformly illuminated image suitable for further processing.

3. Salt-and-Pepper Noise
(a) Interpretation
Salt-and-pepper noise appears as randomly scattered white and black pixels caused by transmission errors or faulty sensors.

(b) Key Issues
- Random black and white dots
- Reduced readability
- Loss of image quality
- Incorrect feature extraction

(c) Filtering Technique
Median Filter

(d) Justification
The Median Filter removes impulse noise while preserving edges better than averaging filters, making it the best choice.

(e) Structured Explanation
Salt-and-pepper noise introduces isolated black and white pixels. A Median Filter replaces each pixel with the median value of its neighbors, effectively removing noise while maintaining important edges.

4. Blurred Edges
(a) Interpretation
After noise removal, the image becomes overly smooth, causing object boundaries to lose sharpness.

(b) Key Issues
- Loss of edge information
- Reduced image sharpness
- Poor object recognition
- Difficulty in feature extraction

(c) Enhancement Technique
Unsharp Masking (Sharpening Filter)

(d) Justification
Unsharp Masking enhances edge details by emphasizing high-frequency components without greatly increasing noise.

(e) Structured Explanation
Excessive smoothing blurs edges. Applying Unsharp Masking restores edge sharpness, improving object visibility while preserving overall image quality.

5. High-Frequency Noise
(a) Interpretation
The satellite image contains fine-grained random noise that hides important image details.

(b) Key Issues
- Reduced clarity
- Difficulty identifying features
- Distorted image texture
- Lower analysis accuracy

(c) Frequency Domain Technique
Low-Pass Filter (Gaussian Low-Pass Filter)

(d) Justification
A Gaussian Low-Pass Filter suppresses high-frequency noise while preserving important low-frequency image information.

(e) Structured Explanation
High-frequency noise affects image quality by introducing unwanted variations. Applying a Gaussian Low-Pass Filter removes this noise and improves image clarity.

6. Object Boundary Detection
(a) Interpretation
Medical images require clear boundaries between tissues for accurate diagnosis and analysis.

(b) Key Issues
- Weak edges
- Noise interference
- Similar tissue intensities
- Boundary discontinuities

(c) Edge Detection Technique
Canny Edge Detection

(d) Justification
The Canny operator provides accurate, thin, and continuous edges while reducing noise using Gaussian smoothing.

(e) Structured Explanation
Boundary detection helps identify tissue structures. The Canny Edge Detector detects edges accurately with minimal noise, making it suitable for medical imaging.

7. Simple Segmentation
(a) Interpretation
Objects and background have clearly different intensity values, making separation straightforward.

(b) Key Issues
- Intensity difference
- Simple object-background separation
- Minimal overlap

(c) Segmentation Technique
Global Thresholding

(d) Justification
Global Thresholding separates foreground and background using a single intensity threshold, making it simple and efficient.

(e) Structured Explanation
Since the object intensity differs clearly from the background, Global Thresholding effectively segments the image by assigning pixels above the threshold to the object and below it to the background.

8. Complex Segmentation
(a) Interpretation
Multiple image regions have overlapping intensity values, making simple thresholding ineffective.

(b) Key Issues
- Intensity overlap
- Similar object appearance
- Complex boundaries
- Noise

(c) Segmentation Technique
Watershed Segmentation

(d) Justification
Watershed Segmentation separates touching and overlapping regions based on image gradients, making it suitable for complex images.

(e) Structured Explanation
Complex segmentation requires advanced methods because intensity values overlap. Watershed Segmentation identifies region boundaries accurately, improving segmentation results.

9. Broken Edges
(a) Interpretation
After edge detection, object boundaries appear incomplete and disconnected.

(b) Key Issues
- Edge discontinuity
- Gaps in boundaries
- Inaccurate object detection
- Reduced segmentation accuracy

(c) Enhancement Technique
Morphological Closing (Dilation followed by Erosion)

(d) Justification
Morphological Closing fills small gaps and connects broken edge segments without significantly changing object shapes.

(e) Structured Explanation
Broken edges reduce object recognition accuracy. Morphological Closing reconnects edge fragments, producing continuous object boundaries.

10. Combined Requirement
(a) Interpretation
An industrial inspection system must remove noise while preserving important edges for accurate defect detection.

(b) Key Issues
- Image noise
- Edge preservation
- Accurate defect identification
- High inspection accuracy

(c) Sequence of Techniques

1. Bilateral Filter for noise removal.
2. Canny Edge Detection for edge extraction.
3. Segmentation for defect detection.

(d) Justification

The Bilateral Filter removes noise while preserving edges. Canny accurately detects edges, and segmentation isolates defects, resulting in reliable industrial inspection.

(e) Structured Explanation

Industrial inspection requires both clean images and sharp edges. First, the Bilateral Filter removes noise while preserving boundaries. Next, Canny Edge Detection extracts important edges, and segmentation identifies defects accurately. This sequence ensures high detection accuracy and reliable inspection results.

11. Low Brightness Image

(a) Interpretation

The image is captured under low-light conditions, making it appear very dark and reducing the visibility of objects.

(b) Key Issues

- Poor visibility
- Loss of image details
- Low contrast
- Difficult object detection
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(c) Enhancement Technique

Brightness Adjustment (Gamma Correction)

(d) Justification

Gamma Correction brightens dark images by increasing pixel intensity while preserving image details better than simple brightness enhancement.

(e) Structured Explanation

A low-brightness image contains insufficient light, making objects difficult to identify. Gamma Correction improves brightness and enhances visibility without significantly affecting image quality.

12. Overexposed Image
(a) Interpretation
The image is excessively bright due to too much light, causing important details to disappear.

(b) Key Issues

- Loss of details in bright regions
- Pixel saturation
- Poor intensity representation
- Reduced image quality

(c) Correction Technique
Histogram Equalization (Brightness Adjustment)

(d) Justification
Histogram Equalization redistributes intensity values, improving contrast and recovering visible details in bright images.

(e) Structured Explanation
Overexposure causes bright regions to lose information. Histogram Equalization improves contrast, making the image more balanced and detailed.

13. Gaussian Noise
(a) Interpretation
Gaussian noise is random noise with a normal distribution that affects smooth regions of an image.

(b) Key Issues

- Grainy appearance
- Reduced image quality
- Hidden fine details
- Poor feature extraction

(c) Filtering Technique
Gaussian Filter

(d) Justification
A Gaussian Filter smooths random noise effectively while preserving the overall image structure.

(e) Structured Explanation
Gaussian noise reduces image quality by adding random intensity variations. Applying a Gaussian Filter smooths the image and improves clarity.

14. Speckle Noise
(a) Interpretation
Speckle noise is multiplicative noise commonly found in ultrasound and radar images.

(b) Key Issues
- Grainy texture
- Reduced image clarity
- Difficult tissue analysis
- Poor edge visibility

(c) Filtering Technique
Lee Filter

(d) Justification
The Lee Filter is specially designed for multiplicative speckle noise and preserves edges while reducing noise.

(e) Structured Explanation
Speckle noise degrades ultrasound images. The Lee Filter effectively suppresses this noise while maintaining important structural information.

15. Loss of Fine Details
(a) Interpretation
Image processing has removed important small features, making the image appear overly smooth.

(b) Key Issues
- Loss of texture
- Reduced sharpness
- Missing small features
- Poor image quality
-

(c) Enhancement Technique
Unsharp Masking

(d) Justification
Unsharp Masking sharpens edges and restores fine image details without significantly increasing noise.

(e) Structured Explanation
Fine details are lost due to excessive smoothing. Unsharp Masking enhances edge sharpness and restores important image features.

16. Edge Enhancement Requirement
(a) Interpretation
The application requires clear edges to improve feature extraction and object identification.

(b) Key Issues

- Weak edges
- Low feature visibility
- Difficult object detection
- Reduced extraction accuracy
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(c) Enhancement Method
Sobel Edge Detection

(d) Justification
The Sobel operator detects horizontal and vertical intensity changes efficiently, highlighting object boundaries.

(e) Structured Explanation
Edge enhancement improves the visibility of object boundaries. The Sobel operator emphasizes intensity changes, making feature extraction easier.

17. Background Noise Removal
(a) Interpretation
Background noise interferes with segmentation, making it difficult to separate objects correctly.

(b) Key Issues

- False segmentation
- Reduced accuracy
- Hidden object boundaries
- Poor image quality

(c) Preprocessing Method
Median Filter

(d) Justification
The Median Filter removes background impulse noise while preserving object edges, improving segmentation accuracy.

(e) Structured Explanation
Background noise affects segmentation performance. Applying a Median Filter removes unwanted noise and produces a cleaner image for accurate segmentation.

18. Threshold Selection Problem

(a) Interpretation

Choosing a single threshold for image segmentation is difficult because intensity values vary across the image.

(b) Key Issues

- Uneven intensity distribution
- Incorrect segmentation
- Poor object separation
- Variable lighting

(c) Thresholding Method

Otsu's Thresholding

(d) Justification

Otsu's method automatically calculates the optimal threshold by minimizing intra-class variance.

(e) Structured Explanation

Manual threshold selection may produce poor segmentation. Otsu's Thresholding automatically determines the best threshold, improving segmentation accuracy.

19. Texture-Based Segmentation

(a) Interpretation

Objects differ mainly in texture rather than pixel intensity.

(b) Key Issues

- Similar intensity values
- Texture variation
- Failure of simple thresholding
- Complex patterns

(c) Segmentation Technique

Texture-Based Segmentation using GLCM (Gray Level Co-occurrence Matrix)

(d) Justification

GLCM extracts texture features such as contrast, homogeneity, and energy, enabling accurate region separation.

(e) Structured Explanation

Intensity-based methods cannot distinguish textured regions effectively. GLCM analyzes texture characteristics, allowing accurate segmentation.

20. Multiple Object Detection
(a) Interpretation
The image contains several objects with similar intensity values, making them difficult to separate.

(b) Key Issues
- Similar object intensities
- Overlapping objects
- Poor segmentation
- Object confusion

(c) Segmentation Technique
Watershed Segmentation

(d) Justification
Watershed Segmentation separates touching or overlapping objects using image gradients, producing accurate boundaries.

(e) Structured Explanation
Objects with similar intensities are difficult to distinguish using simple thresholding. Watershed Segmentation identifies individual object boundaries effectively, improving detection accuracy.

21. Edge Noise Problem
(a) Interpretation
The detected edges contain unwanted noise, making object boundaries unclear.

(b) Key Issues
- False edge detection
- Broken or irregular edges
- Reduced edge accuracy
- Difficulty in object identification

(c) Solution
Gaussian Smoothing followed by Canny Edge Detection

(d) Justification
Gaussian smoothing removes noise before edge detection, and the Canny operator accurately detects true edges while reducing false edges.

(e) Structured Explanation
Noisy edges reduce detection accuracy. Applying Gaussian smoothing first removes noise, and Canny Edge Detection extracts clear and continuous edges for better analysis.

22. Region Merging Requirement
(a) Interpretation
The image contains many small fragmented regions that should be combined into meaningful regions.

(b) Key Issues
- Fragmented regions
- Incomplete segmentation
- Poor object representation
- Increased processing complexity

(c) Technique
Region Merging

(d) Justification
Region Merging combines neighboring regions with similar properties, producing larger and more meaningful segments.

(e) Structured Explanation
Fragmented regions reduce segmentation quality. Region Merging joins similar neighboring regions to improve image interpretation.

23. Over-Segmentation
(a) Interpretation
The segmentation process divides the image into too many small regions.

(b) Key Issues
- Excessive regions
- Increased computation
- Difficult object recognition
- Noise sensitivity

(c) Correction Method
Region Merging (Morphological Closing if needed)

(d) Justification
Region Merging combines similar adjacent regions, reducing unnecessary segmentation and improving object representation.

(e) Structured Explanation
Over-segmentation creates many small regions that complicate analysis. Region Merging combines these regions into meaningful objects for improved segmentation.

24. Under-Segmentation
(a) Interpretation
Different objects are incorrectly merged into one region during segmentation.

(b) Key Issues
- Poor object separation
- Incorrect boundaries
- Reduced segmentation accuracy
- Object confusion

(c) Solution
Watershed Segmentation

(d) Justification
Watershed Segmentation separates connected or merged objects by identifying image boundaries using gradients.

(e) Structured Explanation
Under-segmentation combines multiple objects into one region. Watershed Segmentation separates them accurately, improving segmentation quality.

25. Boundary Refinement
(a) Interpretation
The detected object boundaries are rough, irregular, and not accurately aligned.

(b) Key Issues
- Jagged edges
- Inaccurate boundaries
- Poor object shape representation
- Reduced segmentation quality

(c) Refinement Method
Morphological Operations (Opening and Closing)

(d) Justification
Morphological operations smooth object boundaries, remove small irregularities, and improve boundary accuracy.

(e) Structured Explanation
Rough boundaries reduce segmentation quality. Morphological Opening and Closing refine edges, producing smoother and more accurate object boundaries.

26. Frequency-Based Enhancement
(a) Interpretation
The image requires enhancement using frequency-domain processing instead of spatial-domain techniques.

(b) Key Issues
- Spatial methods may not effectively separate frequency components.
- Noise and useful details exist in different frequency ranges.
- Image quality needs improvement.

(c) Technique
High-Pass Filtering using Fourier Transform

(d) Justification
High-Pass Filtering enhances edges and fine details by emphasizing high-frequency components while suppressing unwanted low-frequency information.

(e) Structured Explanation
Frequency-domain enhancement analyzes the image in terms of frequency components. High-Pass Filtering improves image sharpness and highlights important features.

27. Mixed Noise Scenario
(a) Interpretation
The image contains both Gaussian noise and impulse (salt-and-pepper) noise.

(b) Key Issues
- Two different noise types
- Reduced image quality
- Loss of details
- Difficult filtering

(c) Combined Approach
Median Filter followed by Gaussian Filter

(d) Justification
The Median Filter removes impulse noise, while the Gaussian Filter smooths Gaussian noise, giving better overall image quality.

(e) Structured Explanation
Mixed noise requires multiple filtering techniques. First, the Median Filter removes salt-and-pepper noise, and then the Gaussian Filter reduces Gaussian noise while preserving image quality.

28. Edge Preservation Filtering

(a) Interpretation

The image requires noise removal without losing important edge information.

(b) Key Issues

- Noise removal
- Edge preservation
- Maintaining object boundaries
- Improved image quality

(c) Technique

Bilateral Filter

(d) Justification

The Bilateral Filter removes noise while preserving edges by considering both spatial distance and pixel intensity differences

(e) Structured Explanation

Traditional smoothing filters blur edges. The Bilateral Filter reduces noise while maintaining sharp object boundaries, making it ideal for edge-preserving enhancement.

29. Region Growing Failure

(a) Interpretation

Region Growing fails because incorrect seed points lead to poor segmentation.

(b) Key Issues

- Wrong seed selection
- Incorrect region expansion
- Poor segmentation accuracy
- Missed object boundaries

(c) Correction

Automatic Seed Selection with Region Growing

(d) Justification

Automatic seed selection identifies suitable starting points, leading to more accurate and consistent region growth.

(e) Structured Explanation

Incorrect seed points result in poor segmentation. Automatic Seed Selection improves Region Growing by selecting appropriate initial pixels and producing accurate segmented regions.

30. Industrial Inspection Enhancement

(a) Interpretation

An industrial inspection system must detect defects accurately in noisy images.

(b) Key Issues

- Image noise
- Hidden defects
- Edge preservation
- Accurate segmentation

(c) Enhancement and Segmentation Pipeline

1. **Median/Bilateral Filter** – Remove noise while preserving edges.
2. **Histogram Equalization** – Improve image contrast.
3. **Canny Edge Detection** – Detect object boundaries.
4. **Thresholding or Watershed Segmentation** – Identify defects.

(d) Justification

Noise is removed first to improve image quality. Contrast enhancement makes defects more visible, edge detection identifies boundaries, and segmentation accurately separates defective regions.

(e) Structured Explanation

Industrial inspection requires accurate defect detection in noisy images. A sequence of noise removal, contrast enhancement, edge detection, and segmentation improves image quality and enables reliable defect identification.

Code of Conduct:

I Avinash B(192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: